DynSGS without and with the smoothness cutoff $c = 10^{-3}$

Measured on the smooth Alfvén wave (`problems/MHD/alfvenWaveBabbarRanocha2026`, Babbar & Ranocha 2026, §5.5.1), branch `sm/MHD`, commit `e40bc3e`. The cutoff is the deck entry `AW_CUTOFF` (`:dsgs_cutoff`), 10^{-3} by default.

The coefficient. On element K, with $\Delta_K = \Delta_K^{elem}/(N+1)$ and $w_K = \max(|\mathbf{v}| + c_f)$,

$$\nu_K = \min\big(C_{\max}\,\Delta_K\,w_K,\;\; C_R\,\Delta_K^2\,\max(0,\;r_K - c)\big), \qquad r_K = \max_i \frac{\|R_i\|_{\infty,K}}{n_i}, \qquad R = \partial_t \mathbf{q} + \nabla\cdot\mathbf{F}(\mathbf{q}),$$

R being the element's own strong residual and n_i the domain-wide scale of variable i: its spread $\max_\Omega |q_i - \langle q_i\rangle|$, floored at a scale of the mean state, $\bar\rho$ for ρ, $\bar\rho\bar c$ for the momentum, $\bar\rho\bar c^2$ for E, $\sqrt{\bar\rho}\,\bar c$ for $\mathbf{B}$. Here $\bar\rho = \langle\rho\rangle = 1$ and $\bar c = \sqrt{\gamma\bar p/\bar\rho} = 0.42$ is the sound speed of the mean state ($\bar p = 0.107$, from $\langle\rho E\rangle$, $\langle\rho\mathbf{v}\rangle$, $\langle\mathbf{B}\rangle$). The wave's spread (≤ 0.1) is below that floor, so $n_i = 0.42$ for the momentum and $\mathbf{B}$, and r_K is a rate (1/time): the residual relative to the momentum of a flow at the sound speed. $C_R = 1$, $C_{\max} = 0.5$, $C_{\min} = 0$. **Without the cutoff** is $c = 0$: $\nu_K = C_R\Delta_K^2 r_K$, whatever r_K is. **With it**, $c = 10^{-3}$ is subtracted from the normalized residual first, so $\nu_K = 0$ wherever $r_K \le 10^{-3}$, and ν_K stays continuous across the threshold (it is not an on/off switch).

Without the cutoff: a floor that converges at order 4. The coefficient is computed once per step, and the $\partial_t\mathbf{q}$ in R is the BDF2 of three step-level states, whose truncation error is $\frac{\Delta t^2}{3}\partial_t^3\mathbf{q}$. On a resolved smooth solution the spatial part of R is $O(h^N)$, but a travelling wave has $\partial_t^3\mathbf{q} = \omega^3 A \neq 0$ however well it is resolved ($\omega = 2\pi$, $A = 0.1$). So r_K never falls below

$$r_{\text{floor}} = \frac{\Delta t^2}{3}\,\frac{\omega^3 A}{\bar\rho\bar c} \quad (4\cdot10^{-6} \text{ to } 7\cdot10^{-4} \text{ over the sweep}),$$

and $\nu \approx C_R\Delta^2 r_{\text{floor}} \propto \Delta^2\Delta t^2 \propto h^4$ at a fixed Courant number, *whatever* N. That ν damps the wave by $\nu k^2 t$ ($k = 2\pi$), a state error of $0.14\,\nu k^2 t$:

- it predicts the measured error with no fitted constant ($N = 6$ on 16^2: $5.4\cdot10^{-8}$ both; $N = 5, 6, 7$ within 0.95–1.4× from 8^2 on);
- DynSGS converges at ≈ 4 for $N = 5, 6, 7$, and at equal #DOFs the three curves coincide, since Δ and Δt are both set by the node spacing;
- halving Δt alone divides the error by 3.9–4.0, and the Galerkin error does not move;
- at $N = 3$ the floor is ∼1/100 of Galerkin's own error, and does not show.

With the cutoff $c = 10^{-3}$. $r_{\text{floor}} < 10^{-3}$ on every mesh of the sweep, so $\nu \equiv 0$ and DynSGS *is* the Galerkin scheme: bit for bit, all nine variables, L^2 and L^∞, for $N = 5, 6, 7$ on every mesh but the coarsest of $N = 5$. The orders are Galerkin's (6.5, 6.0, 8.6 on the finest pair). Where the residual is real the cutoff barely matters: at a discontinuity $r_K = O(10^2)$, which c changes by 10^{-5} relative.

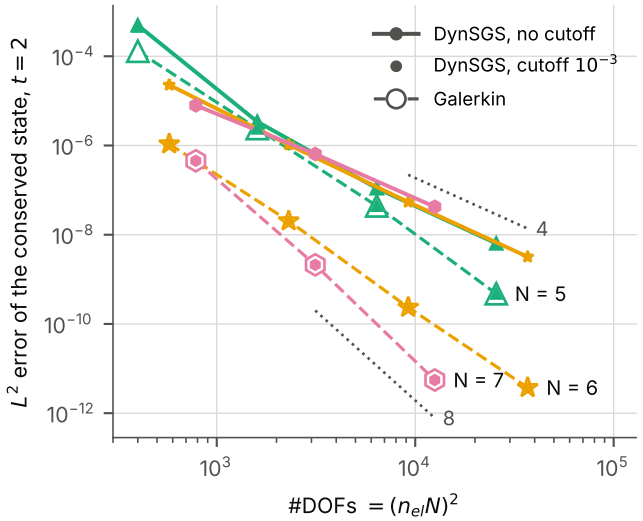


Figure 1: State error at $t = 2$, $N = 5, 6, 7$. Solid: DynSGS without the cutoff (slope 4). Small filled markers inside the hollow Galerkin ones: DynSGS with $c = 10^{-3}$, identical to Galerkin.

N	mesh	no cutoff		$c = 10^{-3}$		Galerkin	
5	8^2	3.4e-6		2.4e-6		2.4e-6	
5	16^2	1.1e-7	5.0	4.4e-8	5.8	4.4e-8	5.8
5	32^2	6.4e-9	4.1	4.8e-10	6.5	4.8e-10	6.5
6	8^2	1.0e-6		2.0e-8		2.0e-8	
6	16^2	5.4e-8	4.3	2.4e-10	6.4	2.4e-10	6.4
6	32^2	3.2e-9	4.1	3.7e-12	6.0	3.7e-12	6.0
7	8^2	6.5e-7		2.1e-9		2.1e-9	
7	16^2	4.2e-8	3.9	5.5e-12	8.6	5.5e-12	8.6

Table 1: State L^2 error and order.

What the cutoff does *not* settle.

- *Under-resolution.* $N = 5$ on 4^2 (20 points per side, the least resolved run): $r_K > 10^{-3}$ in places, ν up to $9\cdot10^{-5}$, error 3.6× Galerkin's. There the viscosity acts, as it should on a marginal wave.
- *Shocks.* $c = 10^{-3}$ is above this wave's floor on every mesh (at most $7\cdot10^{-4}$) and 5 decades below a shock's r_K, but it has not been run on the shock cases (Brio–Wu, Orszag–Tang, rotor, blast). On the smooth MHD vortex ($N = 6$, 32^2, $t = 0.5$) the cutoff likewise gives Galerkin bit for bit; there the excess without it was only 2%.
- *What 10^{-3} means.* c is compared with r_K, the residual *divided by* n_i, so it is a rate in the problem's time units: another normalization (`:dsgs_rel`, element norms, `:dsgs_fast_floors`, which puts the fast speed in place of $\bar c$) or another time scale changes the threshold it stands for.

Reproduce. `tools/alfven_wave_scan.sh` runs the deck as committed, $c = 10^{-3}$, for $N = 3\ldots7$ (`AW_JOBS=5` runs the five orders at once). With `AW_CUTOFF = 0.0` in `user_inputs.jl`, `AW_KEEP=1 AW_VISC=dsgs tools/alfven_wave_scan.sh` adds the curves without the cutoff (records `dsgs_nocut`). `dsgsCutoff_fig.py` redraws Fig. 1 from the case's error store.